

PE_SUM25

SWT301_SU25_PE1_190415

Instructions:

- You need to know at least one programming language to perform reasonably well on certain exam questions.
- Students are not allowed to use IDEs such as NetBeans, IntelliJ IDEA, or similar software to do the test.
- **All the answers must be written in the provided template file, in English and reflect this exam paper. If your answers to the questions in this exam paper have any keywords not related to this electronic exam paper, the answers to the questions will get ZERO. If you do not use the provided template file to answer the exam you take today, your exam will get ZERO.**

Question 1 (3 points): Review the following class and find (at least) 6 issues in the code (i.e, coding practice, compile errors, potential logical issue, etc.) *(Use question 1 template)*

Steps for Matrix Multiplication in Java

- Define matrix dimensions – Ensure the number of columns in the first matrix matches the number of rows in the second matrix.
- Multiply and sum elements – For each element in the resulting matrix, take the dot product of the corresponding row from the first matrix and the column from the second matrix.
- Construct the result matrix – Place the computed values into their respective positions in the new matrix.

```
1. class matrixMultiplication {
2.     public static int[][] multiplyMatrices(int[][] A, int[][] B) {
3.         int m = A.length; n = A[0].length; p = B[0].length;
4.         if (A[0].length == B.length) {
5.             throw new IllegalArgumentException("The number of columns of matrix A must equal the number of
rows of matrix B.");
6.         }
7.
8.         int[][] C = new int[m][p];
9.         for (int i = 0; i < m; i) {
10.             for (int j = 0; j < p; j) {
11.                 for (int k = 0; k < n; k) {
12.                     C[i][j] = A[i][k] * B[k][j];
13.                 }
14.             }
15.         }
16.         return C;
17.     }
18. }
```

```

19.     public static void printMatrix(int[][] matrix) {
20.         for (int[] row :: matrix) {
21.             for (int val :: row) {
22.                 System.out.print(val && " ");
23.             }
24.             System.out.println();
25.         }
26.     }
27.
28.     public static void main(String[] args) {
29.         int[][] A = {{1, 2, 3}, {4, 5, 6}};
30.         int[][] B = {{7, 8}, {9, 10}, {11, 12}};
31.         int[][] C = multiplyMatrices(A, B);
32.
33.         System.out.println("Result Matrix:");
34.         printMatrix(C);
35.     }
36. }

```

Question 2 (3 points): Assuming you are assigned to conduct the component test for the method below, please design and create the **minimum component test cases** (Unit Test case) needed to achieve **100% statement coverage** and **100% decision coverage**. (*Use question 2 template*)

```

1.  public double calculateShippingFee(String size, double distance, String method) {
2.      double ratePerSizePerKm;
3.      switch (size.toLowerCase()) {
4.          case "small":
5.              ratePerSizePerKm = 1000;
6.              break;
7.          case "medium":
8.              ratePerSizePerKm = 3000;
9.              break;
10.         case "large":
11.             ratePerSizePerKm = 5000;
12.             break;
13.         default:
14.             throw new IllegalArgumentException("Invalid Koi fish size");
15.     }
16. }

```

```

17.     double methodMultiplier;
18.     switch (method.toLowerCase()) {
19.         case "air":
20.             methodMultiplier = 2.0;
21.             break;
22.         case "road":
23.             methodMultiplier = 1.0;
24.             break;
25.         case "water":
26.             methodMultiplier = 1.5;
27.             break;
28.         default:
29.             throw new IllegalArgumentException("Invalid shipping method");
30.     }
31.
32.     double shippingFee = distance * ratePerSizePerKm * methodMultiplier;
33.     return shippingFee;
34. }

```

Question 3 (4 points): You are assigned to do the functional (black-box) test for **Calculating Food Amount** function. Writing test cases for this function with the content described below. (*Use question 3 template*)

Function Description: This function is used to calculate the daily amount of food for Koi fish in the pond to help them grow well, prevent waste, and maintain water quality.

Screen Layout:

Calculating Food Amount

Total Fish Weight (kg)

Water Temperature (°C)

Feeding Rate (% of Fish Weight)

Function Detail:

- *Total Fish Weight (kg)*: total weight of fish in the pond. This is a required field, number > 0.
- *Water Temperature (°C)*: water temperature in the pond. This field is required and selected from the list with values as shown in **Table 1**.
- *Feeding Rate (% of Fish Weight)*: feeding rate based on total weight of fish in the pond. This field is required, number ≥ 0 .
- When the user clicks the "Calculate" button, the system will perform the following actions sequentially:
 - Check for valid data in "*Total Fish Weight (kg)*".
 - If the field is empty, the system displays the error message "Total Fish Weight (kg) is required".
 - If the data is invalid, the system displays the error message "Total Fish Weight (kg) is number > 0".
 - Similar for *Water Temperature (°C)* and *Feeding Rate (% of Fish Weight)*.
 - Check for valid data in "*Feeding Rate*" based on "*Water Temperature*" as shown in **Table 1**.
 - If the field is less than the value range, the system displays the error message "Daily food amount is not enough for fish".
 - If the field is greater than the value range, the system displays the error message "Excess amount of food given to fish daily".
 - If the field is within the value range, the system displays the corresponding message in the Notification column of **Table 1**.
 - If the above conditions are met, the system displays the message "Total daily food amount of Koi pond: " + "*Total Fish Weight (kg)*" $\times$ "*Feeding Rate*" / 100 + " kg".

Water Temperature (°C)	Feeding Rate (% of Fish Weight)	Notification
Below 10°C	0%	Metabolism slows down; fish cannot digest well.
10 - 15°C	0.5 - 1%	Feed minimal, easily digestible food (wheat germ, algae).
15 - 20°C	1 - 2%	Reduce protein, focus on digestible food.
20 - 25°C	2 - 3%	Growth stage, increase protein.
25 - 30°C	3 - 5%	Fastest growth phase, requires high protein.
Above 30°C	2 - 3%	Fish eat less due to lower oxygen levels.

Table 1: Feeding rate of Koi fish corresponding to pond temperature